University of Michigan

NERS 515

Nuclear Measurements Laboratory

Laboratory 1: An Introduction to Lab Equipment and Methods

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≥10 pgs, needs table of contents

Abstract

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This laboratory introduces fundamental nuclear measurement instrumentation, specifically the oscilloscope, pulse generator, counter, and linear amplifier, to characterize simulated radiation detector signals. A pulse generator simulated detector outputs, allowing for the precise measurement of signal amplitude, period, and characteristic rise and fall times using an oscilloscope. The signal frequency was verified using a counter, and the observed statistical errors aligned perfectly with the theoretical Poisson counting statistics. Furthermore, coaxial cable reflections were examined under various termination loads. Terminating the line with its 50 Ω characteristic impedance successfully minimized signal reflection, whereas open and short circuits maximized distortion. A measured round-trip time delay of 131.44 ns was also utilized to calculate the length of the physical coaxial cable at approximately 13.01 m. Finally, pulse-shaping analysis confirmed that the linear amplifier maintains a highly linear response ($R^2 = 0.99755$) up to a saturation threshold of 12 V. These results establish a foundational understanding of signal processing techniques and the equipment limitations necessary for accurate nuclear measurements.

good

uncertainty?

1 Introduction

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Next time try to write the intro in terms of applicability to radiation detection, not specifically to 515

The laboratory experiments in NERS 515 (Nuclear Measurements Laboratory) require working knowledge of various instrumentation, the theory behind why they work, and statistical concepts relevant to experimental data analysis [1]. Four common devices we are interested in are oscilloscopes, pulsers, counters/timers, and linear amplifiers. Oscilloscopes visualize input signals and provide tools to precisely measure the shape, size and time-dependent features, pulsers generate these input signals, counters track the number of pulses within an interval, and linear amplifiers is a common tool that shapes the signal [1]. It is also important to understand how the properties of coaxial cables we use to send signals between machines effect the output information (e.g. cable reflection).

and amplifies it!

This initial laboratory is an introduction to the instrumentation and methods pertaining to nuclear measurements, as well as standard laboratory habits for record keeping and reports and habits useful to the course and beyond.

okay

2 Theory

Since this lab is to introduce us to lab equipment and standard procedures, understanding the physics and theory behind how they work is important.

Oscilloscopes visualize one or more input signals and provide tools to measure properties of the signal, which in our case is a pulse [1]. Our main use for the oscilloscope is to verify that the system is responding as expected by measuring their amplitudes and calculate other properties such as frequencies and rise and fall times. It captures this waveform by sampling the input at a fixed clock frequency and digitizing each sample with an ADC [2]. The sampling frequency must be at least twice the highest frequency component in the signal (the Nyquist rate) to avoid signal distortion.

For this lab, we observe a pulse generated by a pulser, which simulates the electronic response of radiation detectors [1]. A tail pulse generator provides independently adjustable rise and fall times, mimicking the tail-pulse output of a preamplifier connected to a real radiation detector, where the fast leading edge corresponds to the detector's charge collection time and the long decay tail reflects the time constant of the collection circuit [2].

Counters or timers count the number of pulses within a chosen time interval, incrementing with each incoming pulse [2].

Linear amplifiers is a pulse-shaping tool that amplifies the signal linearly, multiplying the input pulse by gain set by the user [2]. This is often used to bring small preamplifier outputs to a more suitable range. The amplification must be linear and proportional to the input. Otherwise, different energies would scale by different factors, distorting the original pulse.

Coaxial cables consist of a central conductor sheathed by a dielectric insulator with an outer shield that blocks external electromagnetic interference [2]. These cables are used because its characteristic impedance Z_0 does not depend on the cable length, but the dielectric and conductor geometry, and also because the outer shield minimizes noise pickup. Z_0 is the ratio of the applied step voltage over the current that the cable draws while the voltage propagates through it. As a result, a current is reflected which can be shown in an oscilloscope. If the cable is terminated with its characteristic impedance, the termination draws the same current as when the step arrives. Nothing changes at the boundary and there is no reflection. If the cable is floating (left open), the reflected pulse is of the same polarity and amplitude of the original current. If the the cable is shorted, it reflects a pulse of opposite polarity but equal magnitude.

8/10

whose?

what is the system?

define acronym the first time you use it

good

great evaluation!

exactly!

good

Remember to introduce any equations you will need in the theory section

3/3

3 Equipment and Procedure

Details of all equipment used in this lab are listed in Table 1.

Purpose	Manufacturer	Model	Serial Number
Oscilloscope	Agilent Technologies	DSO6052A	MY44007040
Tail Pulse Generator	BNC	BH-1	32412
Counter/Timer	Ortec	994	2108
Amplifier	Ortec	572A	121

good

Table 1: List of equipment used in Laboratory 1.

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3.1 Procedure

Can probably shorten this for future reports

For each experiment we mostly followed the procedure described in the Fall 2026 Experiment 1 Laboratory Manual [1], additionally taking screenshots of the waveform and measurements to reference in later calculations and reports. We assumed the error to be half of the measured line width of the waveform. Our procedure deviated from that of the manual for the amplifier linearity experiment due to a misreading of the instructions.

⇒good!

⇒the modifications we made can be found in Section...

3.1.1 Oscilloscope Measurements

We set up our oscilloscope and pulser according to the block diagram in Figure 1a with a rise time 0.5 μs, fall time 5 μs, input impedance 1 MΩ, and amplitude 1 V. Using the cursors, we measured the baseline and maximum value of the pulse and their uncertainties to obtain the amplitude and frequency. For the rise time, we measured the time it takes for the pulse to go from 10% to 90% and 37% to 100% and the reverse for fall time. To verify that the sampling frequency is above the Nyquist rate, we measured the period between each count of data to calculate the frequency.

good

To check the the calibration of the scope, we setup the equipment following Figure 1b where we measure the amplitude and frequency of the produced square wave to be 2.5 V and 1.2 kHz respectively.

To verify the frequency measured by the scope, we use a counter to count the number of pulses within a fixed time interval and compare it to that measured by the scope. The setup for this experiment follows Figure 1c.

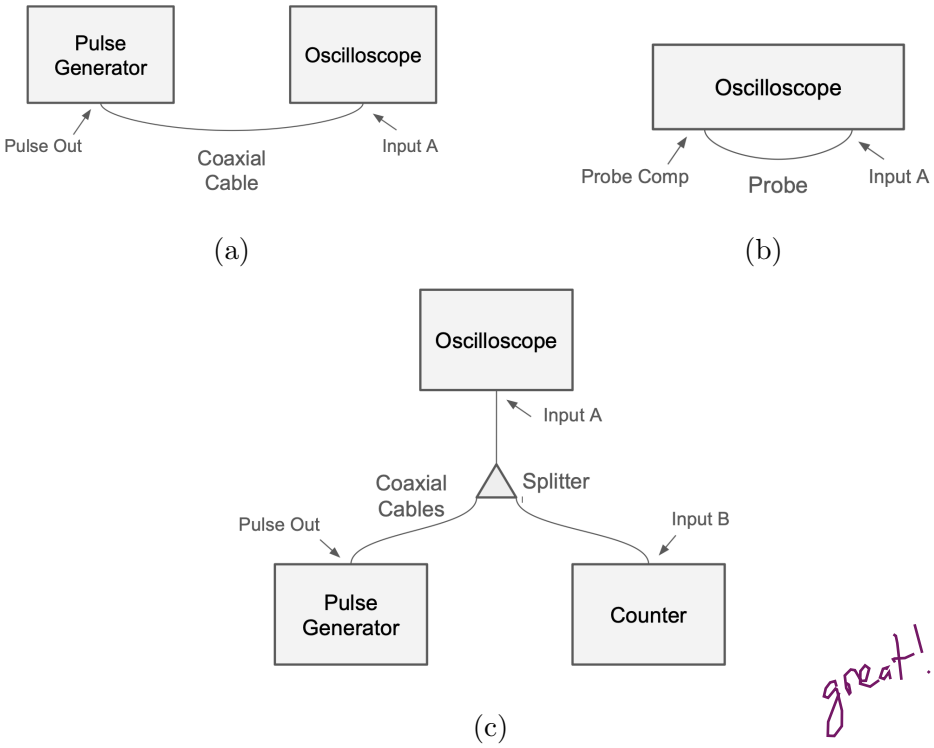


Figure 1: Equipment setup block diagrams for oscilloscope measurement experiments.

3.1.2 Cable Reflections

In this part of the experiment, we observe how coaxial cable reflections affect the shape of the pulse.

We can calculate the cable length by measuring the time interval between the initial and reflected pulse with the setup shown in Figure 2, using a splitter to reflect the pulse down a 50 ft coaxial cable. The pulse rise and fall time are set to their minimum values of 0.02 μs and 0.05 μs, respectively. While we know the cable length, we can use Table 16.1 in [2] to calculate it with the cable's propagation velocity.

To observe how different terminations affect the reflected pulse, we use the same setup but terminate the long coaxial cable in four different ways: shorted with a paper clip, floating (not connected to anything), with the characteristic impedance 50 Ω, and an impedance of 75 Ω.

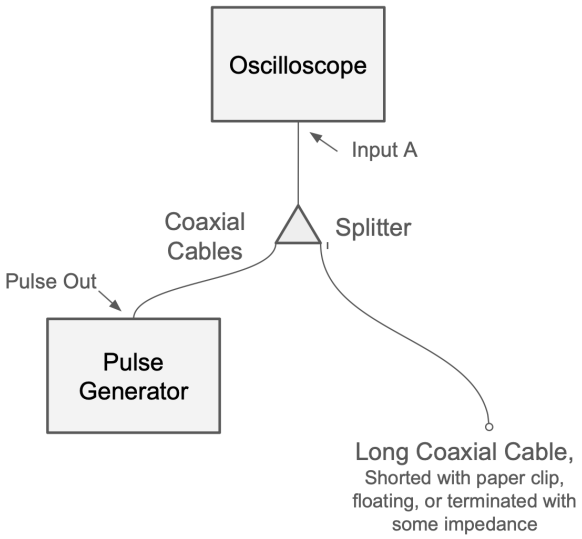


Figure 2: Equipment setup block diagram for cable reflection experiments.

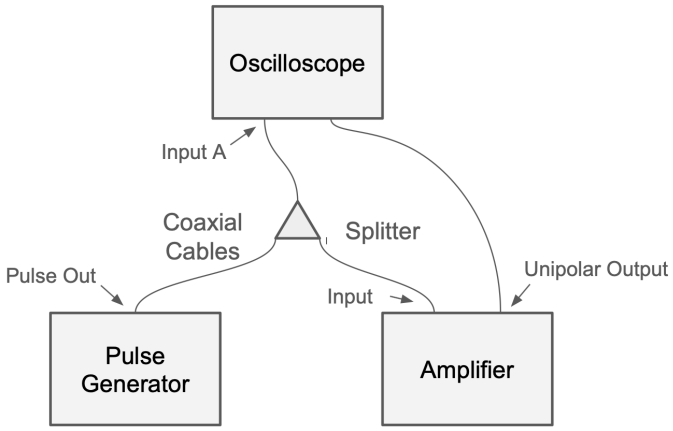


Figure 3: Equipment setup block diagram for amplifier experiments.

3.1.3 Amplifier Linearity

We also verify the linearity of our pulse amplifier, otherwise unwanted distortions to our original pulse occur. Following the setup shown in Figure ??, we connect our pulser, scope, and amplifier using a splitter. The rise time is 1 μs, fall time is 100 μs, and the amplitude is set to 0.1 V. The gain of

the amplifier is initially set so that the amplitudes shown in the scope are 1 V. We observe the saturation voltage of the amplifier by increasing the gain until the it saturates while keeping all other settings constant.

For this section, we were also instructed to observe how the amplifier affects five different output pulses by changing the output amplitude of the *pulser* while leaving all amplifier settings constant, including the gain. We instead measured the amplitudes for five different amplifier gains while keeping the pulser settings unchanged.

4 Data

Measurement	Value	Errors
Amplitude	169.375 mV	12.357 mV
Period	6.713 ms	0.838 ms

Table 2: Summary of initial oscilloscope measurements for amplitude and period, including combined base and peak errors.

Measurement	Threshold / Interval	Calculated Value
Rise Time	10% – 90%	352.600 ns
Rise Time	37% – 100%	684.0 ns
Fall Time	10% – 90%	11.225 μs
Fall Time	37% – 100%	5.405 μs

Table 3: Calculated time intervals for evaluating the rise and fall time characteristics of the signal throughout different locations of the signal's curve.

Parameter	Value
Period	12.500 ns
Frequency	80 MHz

Table 4: Sampling frequency parameters.

Raw, unprocessed data should be in this section, but most values given are processed, so they belong in analysis.

Parameter	Value	Errors
Amplitude	2.5 V	142.908 mV
Frequency	1.198 kHz = 834.794 ns	115.82 ns

these are not the same thing

Table 5: Probe analysis measurements including amplitude and frequency, with calculated baseline and peak errors.

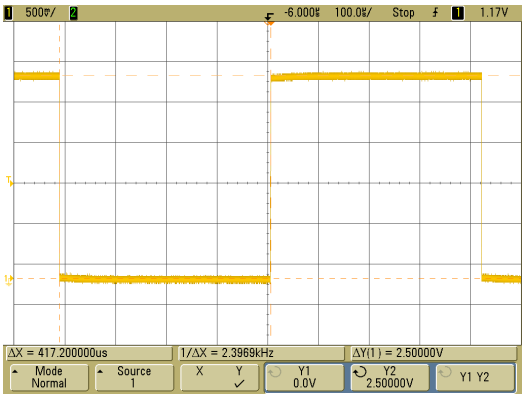


Figure 4: Probe Calibration Peaks shown in Table 5.

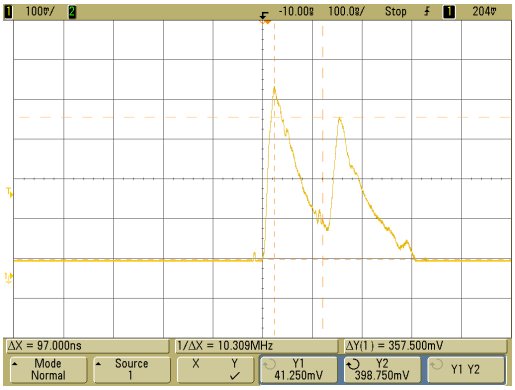
Parameter	Value	Peak 1 Error	Peak 2 Error
Counts in 1 minute	89263 ± 298.769	–	–
Period	672.228 μs	3.6 μs	3.58 μs
Frequency	1.4876 kHz	–	–

Table 6: Frequency verification data based on 1-minute counts, including period and frequency error analysis.

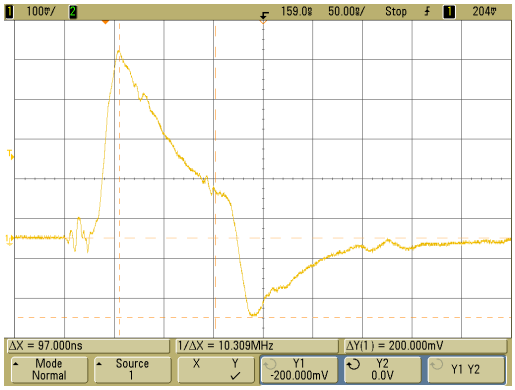
Termination Load	Reflected Amplitude	Base Error (Δy_b)	Peak Error (Δy_p)
Floating	357.5 mV	20 mV	13.75 mV
Shorted	200 mV	20 mV	15 mV
Some Other Impedance (75 Ω)	153.75 mV	20 mV	18.75 mV
Characteristic Impedance (50 Ω)	85 mV	20 mV	25 mV

Note: Error Peak 1 = 5.18 ns; Error Ref. Peak = 5.32 ns. for what?

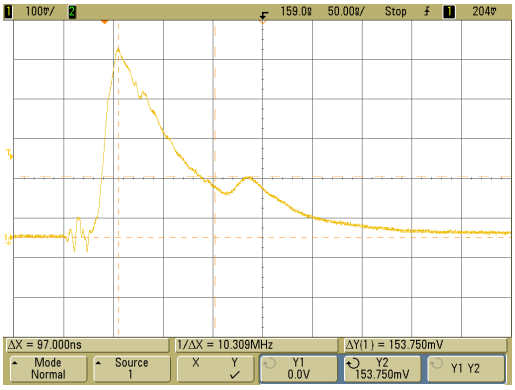
Table 7: Reflected signal amplitudes and associated baseline/peak errors for a 50 Ω characteristic impedance coaxial cable under various termination load conditions (floating, shorted, 75 Ω, and 50 Ω).



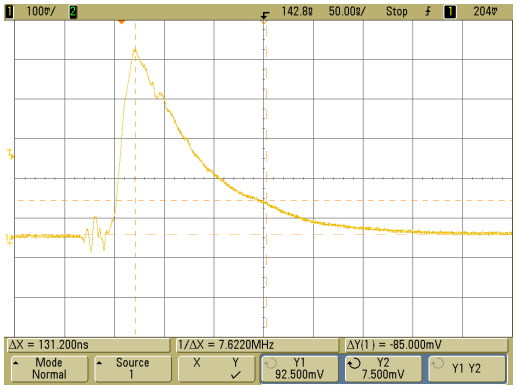
(a) Floating Cable Amplitude.



(b) Shorted Cable Amplitude.



(c) Other Impedance Amplitude (75Ω).



(d) Characteristic Amplitude (50Ω).

good

Figure 5: Oscilloscope readings of reflected pulses from different terminations based on Table 7.

Gain	Shaped Amplitude	Base Error (Δy_b)	Peak Error (Δy_p)
5.8	1.480 V	23.8 mV	28.12 mV
13.8	1.950 V	23.8 mV	28.12 mV
34.5	4.738 V	23.8 mV	28.12 mV
74.0	10.12 V	23.8 mV	28.12 mV
91.2	12 V	23.8 mV	28.12 mV

Table 8: Shaped pulse amplitude measurements of a 0.1 V pulse subject to various amplifier gains.

uncertainty?

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5 Analysis of Data

5.1 Basic Oscilloscope Readings

As outlined in the theoretical framework, the pulser simulates the electronic response of a radiation detector, and the oscilloscope is utilized to visualize this input signal and measure its time-dependent features. The initial oscilloscope readings yielded a pulse amplitude of 169.375 mV with a propagated uncertainty of 12.357 mV, alongside a period of 6.713 ms with an error of 0.838 ms (Table 2).

Sample calculations?

By tracking the signal's shape over time, the measured rise time from 10% to 90% amplitude was found to be 352.600 ns, which expanded to 684.0 ns when evaluating the 37% to 100% interval. The fall times were significantly longer, measuring 11.225 μs for the 10% to 90% interval and 5.405 μs for the 37% to 100% interval (Table 3).

How is this significant? What does it mean for using the equipment? How does it compare to the settings?

5.2 Error Propagation

To determine the total uncertainty for measurements derived from two distinct points on an oscilloscope trace, the individual errors were combined into a single, average error value. For amplitude measurements, the total uncertainty (ΔY)is calculated from the baseline error (Δy_b) and the peak error (Δy_p) using the following equation [2]:

Good!

$$\Delta Y = \sqrt{(\Delta y_b)^2 + (\Delta y_p)^2}$$
$$\Delta Y = \sqrt{8.75^2 + 8.125^2}$$
$$\Delta Y = 12.357$$

units?

Similarly, the total uncertainty for the period and frequency measurements (ΔX) is determined by combining the independent horizontal peak errors (Δx_1 and Δx_2):

$$\Delta X = \sqrt{(\Delta x_1)^2 + (\Delta x_2)^2}$$
$$\Delta X = \sqrt{0.576^2 + 0.608^2}$$
$$\Delta X = 0.838$$ units?

These equations were applied to calculate the combined uncertainties reported in the initial oscilloscope measurements, in Table 2.

5.3 Probe Measurements

During the probe calibration, the theoretical expectation of a steady reference signal was verified. The square wave amplitude was measured at 2.5 V with a combined error of 142.908 mV. The corresponding period was 834.794 ns (1.198 kHZ) with an uncertainty of 115.82 ns.

5.4 Frequency Verification and Counting Statistics

The theory section establishes that counters track the number of discrete pulses within a fixed time interval. To verify the oscilloscope's calculated frequency, pulses were sent to the counter over a 1-minute interval, resulting in $N = 89,263$ counts. For nuclear radiation measurements, random pulse generation follows a Poisson distribution, where the standard deviation (σ_N) is defined as the square root of the number of counts [2]:

true, but that's not what this is. So why can you use Poisson?

$$\sigma_N = \sqrt{N} = \sqrt{89263} \approx 298.77$$

This theoretical calculation aligns exactly with the experimental counting error of ± 298.769 reported in the verification data. The corresponding measured frequency evaluated during this interval was 1.4876 kHz. ← uncertainty?

5.5 Cable Reflection Analysis

While coaxial cables are necessary to transport information between instrumentation, their physical and electrical properties can cause reflections that

affect the output signal. The cable reflection experiment demonstrated how different termination loads (Z_L) influence this distortion. The theoretical reflection coefficient Γ dictates the fraction of the voltage reflected and is governed by the equation [2]:

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$$

where Z_0 is the characteristic impedance of the coaxial cable that was used (50 Ω) [1].

In addition to analyzing the reflection amplitudes, the physical length of the cable (L) was determined using the time interval between the initial and reflected pulses. The measured round-trip time delay (Δt) for the setup was 131.44 ns. According to Knoll [2], standard coaxial cables with a solid polyethylene dielectric have a propagation velocity of approximately $v \approx 0.66c$. The length is calculated using the kinematics equation for a round-trip distance:

$$L = \frac{v\Delta t}{2}$$

$$L = \frac{(0.66 \times 3 \times 10^8 \text{ m/s})(131.44 \times 10^{-9} \text{ s})}{2}$$

$$L \approx 13.01 \text{ m}$$

The measured time delay (Δt) represents the round-trip transit time of the signal. The pulse travels down the entire length of the cable (L), reflects at the termination boundary, and travels the entire length back to the oscilloscope (an additional distance L). Because the total distance traveled is $2L$, we can apply the fundamental kinematic equation for distance ($d = vt$) to yield $2L = v\Delta t$. Rearranging this equation to solve for the cable length gives $L = \frac{v\Delta t}{2}$ [2].

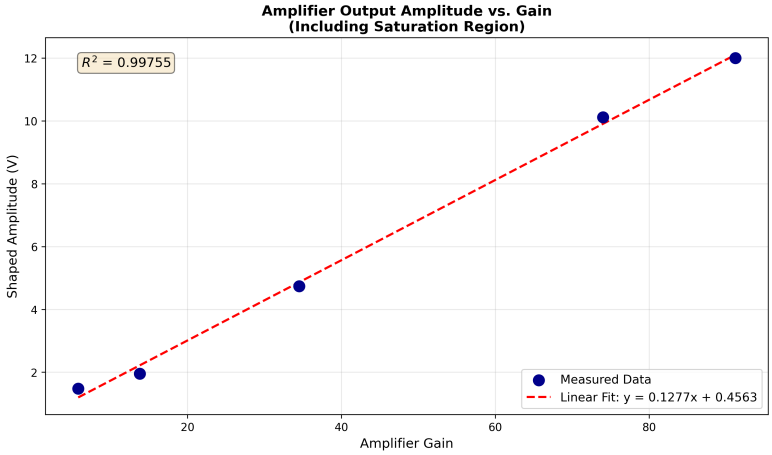
- **Floating (Open Circuit):** When the cable is floating ($Z_L \to \infty$), the reflection coefficient approaches $\Gamma = 1$. This theoretical maximum reflection aligns with the large reflected amplitude of 357.5 mV observed during the experiment.

- **Shorted Circuit:** A shorted cable ($Z_L = 0$) theoretically yields $\Gamma = -1$, causing the pulse to reflect entirely with inverted polarity. The magnitude of the measured reflection was 200 mV.

- **Intermediate Impedance:** Terminating the cable with a 75 Ω impedance resulted in a smaller reflected amplitude of 153.75 mV.
- **Characteristic Impedance:** Terminating the cable with an impedance matching its characteristic impedance ($Z_L = 50\ \Omega$) results in a theoretical $\Gamma = 0$, which should eliminate reflections. The experimental data confirms this theoretical principle, showing the smallest measurable response (85 mV) under the 50 Ω termination condition.

how do you get this?

5.6 Amplifier Linearity



Error bars?

Figure 6: Resulting shaped amplitude from various amplifier gains from Table 8.

The data from Table 8 is plotted in Figure 6 and fitted linearly. The calculated correlation $R^2 = 0.99755$, which suggests that the amplifier gain is linear in the gain setting, but it does not prove that the amplifier itself is linear across various input values. At a gain of 100.912 on the 0.1 mV input pulse, the output pulse began to saturate at its limit 12 V.

uncertainty?

6 Conclusions

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This laboratory successfully demonstrated the fundamental operation of standard nuclear measurement instrumentation. Oscilloscope measurements ef-

fectively characterized the time-dependent features of a simulated detector pulse. Frequency verification using a counter confirmed that the experimental statistical error aligned perfectly with theoretical Poisson counting statistics. Furthermore, the coaxial cable experiment validated that terminating a circuit with its characteristic impedance (50 Ω) successfully minimizes signal reflection.

why do these matter to our work?

Analysis of the linear amplifier yielded a strong linear relationship between gain settings and output amplitude ($R^2 = 0.99755$). However, this procedure deviated from the laboratory manual, which required varying the input amplitude at a constant gain to verify true input-output linearity.

The amplifier exhibited a nearly linear response ($R^2 = 0.99755$) as long as the voltage remained under the 12 V saturation point. To ensure accurate signal processing without distortion, the amplifier must operate below this threshold. good

References

[1] University of Michigan, Department of Nuclear Engineering and Radiological Sciences, *NERS 515 Nuclear Measurements Laboratory: Laboratory 1*, 2026.

[2] G. F. Knoll, *Radiation Detection and Measurement*. Hoboken, NJ: John Wiley & Sons, Inc., fourth ed., 2010.

General format

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